

Proof of baseline invariance

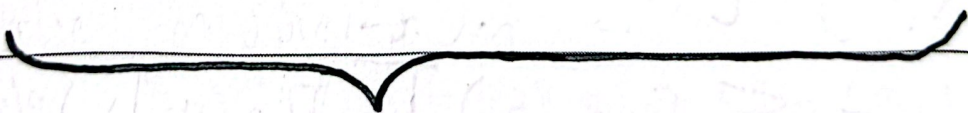
Consider $\nabla_{\theta} J(\theta) = E_{\tau \sim \pi_{\theta}} [R(\tau) \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$

Consider subtracting a baseline $b(s_t)$ depending only on states... the change is equivalent to subtracting ① below from the above.

$$E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T b(s_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \dots \textcircled{1}$$

We will prove $\textcircled{1} = 0$.

$$\textcircled{1} = \sum_{t=0}^T E_{\tau \sim \pi_{\theta}} [b(s_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$$


 $\textcircled{2}$

Consider the contribution of (2) for any fixed $t=k$.

$$E_{\tau \sim \pi_\theta} [b(s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k)] \dots \textcircled{3}$$

$$\textcircled{3} = \sum_{\tau} P(\tau) b(s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k)$$

$$= \sum_{s_k} \sum_{a_k} \sum_{f^k \dots} P(\tau) b(s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k)$$

Where f^k enumerates over all subsequences following s_k .

$$= \sum_{s_k} d_k^{\pi}(s_k) b(s_k) \sum_{a_k} \pi_{\theta}(a_k | s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k) \sum_{f^k \dots} \text{Prob}(f^k \dots | s_k, a_k)$$

$d_k^{\pi}(s_k)$ is a "state distribution, collapsing the probability of reaching a state s_k .

$\sum_{f^k \dots} \text{Prob}(f^k \dots | s_k, a_k)$ sums to 1, as

this sums over all conditional probabilities of subsequences given s_k, a_k .

Then $\sum_{a_k} \pi_{\theta}(a_k | s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k) = \sum_{a_k} \nabla_{\theta} \pi_{\theta}(a_k | s_k)$

by the log-derivative trick.

$$\sum_{a_k} \nabla_{\theta} \pi_{\theta}(a_k | s_k) = \nabla_{\theta} \sum_{a_k} \pi_{\theta}(a_k | s_k) = \nabla_{\theta} 1 = 0$$

Thus, the middle sum $\sum_{a_k} \pi_{\theta}(a_k | s_k) \nabla_{\theta} \log \pi_{\theta}(a_k | s_k)$ is 0 for any fixed $t=k$.

Thus, the contribution of any ③ is 0, so ② is 0, and thus ① = 0.

Thus, subtracting a baseline $b(s_t)$ from a trajectory level reward $R(\tau)$ in the vanilla policy gradient does NOT change the gradient $\nabla_{\theta} J(\theta)$.